

`/* Question`

`1a) Data Types:`

Write a Java program to store and display the following details of a household:

Number of family members - integer

Water consumed in litres - decimal value

House number - integer

Water usage status - character

Use appropriate Java data types for each value and display all the details.*/

`//Answer`

`import java.util.Scanner;`

`public class Household {`

`public static void main(String[] args) {`

`Scanner sc = new Scanner(System.in);`

`int familyMembers;`

`double waterConsumed;`

`int houseNumber;`

`char usageStatus;`

`System.out.print("Enter number of family members: ");`

`familyMembers = sc.nextInt();`

`System.out.print("Enter water consumed in litres: ");`

`waterConsumed = sc.nextDouble();`

`System.out.print("Enter house number: ");`

`houseNumber = sc.nextInt();`

`System.out.print("Enter water usage status (H/L): ");`

`usageStatus = sc.next().charAt(0);`

`System.out.println("\n--- Household Details ---");`

`System.out.println("Family Members: " + familyMembers);`

`System.out.println("Water Consumed: " + waterConsumed + " litres");`

`System.out.println("House Number: " + houseNumber);`

`System.out.println("Water Usage Status: " + usageStatus);`

`}`

`}`

/* 1b) If-Else Condition:

Write a Java program to calculate the water bill based on water consumption. Read the water consumption in litres.

If consumption is 500 litres or less, the bill is Rs.100.
If consumption is more than 500 litres, the bill is Rs.200.
Use an if-else statement and display the water bill.*/

```
//Answer
import java.util.Scanner;
public class WaterBill {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.print("Enter water consumption in litres: ");
double consumption = sc.nextDouble();
if (consumption <= 500) {
System.out.println("Water Bill = Rs.100");
}
else { System.out.println("Water Bill = Rs.200");
}
}
}
```

/* 1c) Methods:

Write a Java program to calculate the total water consumption of a household using a method.

Create the following method:

calculateTotal(int morningUsage, int eveningUsage)

The method should return the total water consumption. Read the morning and evening water usage from the user, call the method, and display the total consumption*/

```
//Answer
import java.util.Scanner;
public class WaterConsumption {
    static int calculateTotal(int morningUsage, int eveningUsage) {
        return morningUsage + eveningUsage;
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter morning water usage: ");
        int morningUsage = sc.nextInt();
        System.out.print("Enter evening water usage: ");
        int eveningUsage = sc.nextInt();
        int total = calculateTotal(morningUsage, eveningUsage);
        System.out.println("Total water consumption: " + total + " litres");
    }
}
```

```
C:\Users\hp\OneDrive\java_lab>javac WaterBill.java
```

```
C:\Users\hp\OneDrive\java_lab>java WaterBill
```

```
Enter water consumption in litres: 8000
```

```
Water Bill = Rs.200
```

```
C:\Users\hp\OneDrive\java_lab>javac WaterConsumption.java
```

```
C:\Users\hp\OneDrive\java_lab>java WaterConsumption
```

```
Enter morning water usage: 78
```

```
Enter evening water usage: 90
```

```
Total water consumption: 168 litres
```

```
C:\Users\hp\OneDrive\java_lab>javac HouseHold.java
```

```
C:\Users\hp\OneDrive\java_lab>java HouseHold
```

```
Enter number of family members: 6
```

```
Enter water consumed in litres: 99
```

```
Enter house number: 3456
```

```
Enter water usage status (H/L): 20
```

```
--- Household Details ---
```

```
Family Members: 6
```

```
Water Consumed: 99.0 litres
```

```
House Number: 3456
```

```
Water Usage Status: 2
```

```
C:\Users\hp\OneDrive\java_lab>|
```